

CareerForge – AI-Powered Career Development Platform with DevOps, DevSecOps & MLOps Pipeline

Team Members :

S.NO	NAME	ROLL NUMBER
1	Somu Vaishnavi	2420030024
2	T. Dhanya Sri	2420030338
3	M. Srihitha	2420030405
4	Bhagyalaxmi	2420030700

Faculty Guide : Ms. G Lavanya

Abstract

CareerForge – AI-Powered Career Development Platform with DevOps, DevSecOps & MLOps Pipeline is a web-based platform designed to help students improve career readiness through AI-powered resume analysis, skill-gap identification, personalized learning recommendations, and interview preparation. The system provides secure authentication, user profiles, progress tracking, and an interactive career dashboard. Developed using Agile and Scrum practices, the platform integrates React.js, Spring Boot, and PostgreSQL. DevOps practices use Docker, GitHub Actions, and AWS for CI/CD and cloud deployment, while DevSecOps incorporates secure authentication and vulnerability scanning. Python, Scikit-learn, and MLflow support the machine learning lifecycle, creating a secure, scalable, and intelligent career development solution.

Problem Statement

Students often struggle with:

- Resume improvement
- Identifying skill gaps
- Choosing suitable career paths
- Finding relevant learning resources
- Preparing for technical interviews

Existing Challenge

Career preparation is often fragmented across multiple platforms, making it difficult for students to receive personalized and continuous guidance.

Objectives

- Develop an AI-powered career development platform
- Analyze resumes and identify relevant skills
- Identify skill gaps for target career roles
- Provide personalized career and learning recommendations
- Support interview preparation
- Track student career progress
- Apply Agile, DevOps, DevSecOps and MLOps practices

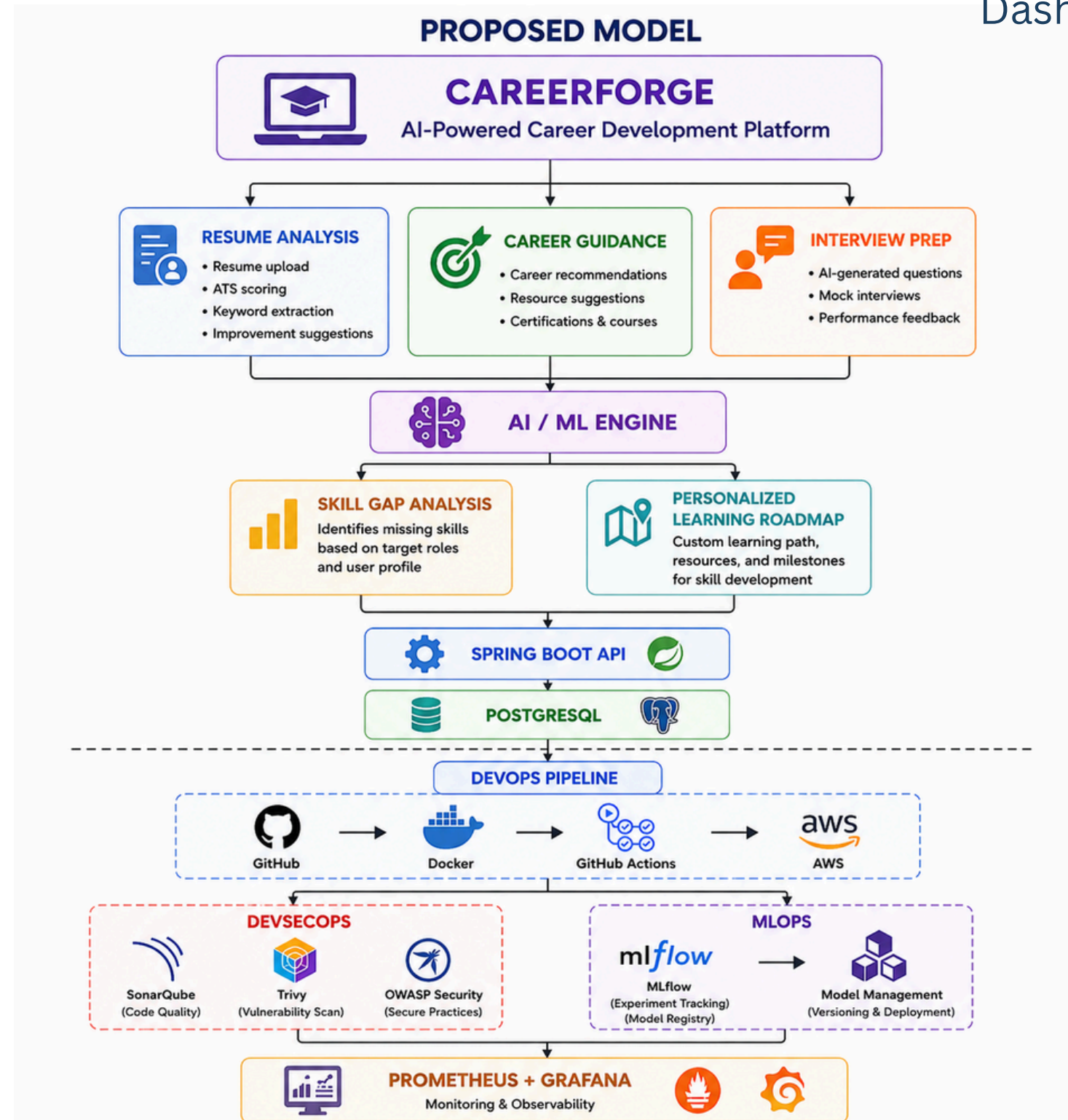
Existing Model

Existing System	Main Features	Limitation / Gap
<u>Jobscan</u>	ATS resume analysis, keyword matching, resume optimization	Mainly focused on resume/ATS optimization
<u>Teal</u>	AI resume builder, job matching, resume analysis	Strong career/job-search tools but limited DevOps/MLOps focus
<u>Career.io</u>	Resume optimization, job tracking, interview preparation	Broad career platform but doesn't focus on an integrated academic DevOps/DevSecOps/MLOps pipeline
<u>LinkedIn Career</u>	Job search, career guidance, AI-assisted career and resume features	Primarily a professional networking and recruitment platform
AI Career Recommendation Systems	Resume analysis, job-role prediction, skill-gap analysis, learning recommendations	Often research prototypes rather than complete production-oriented platforms

Proposed Model

CareerForge integrates career guidance, resume analysis, interview preparation, AI/ML, DevOps, DevSecOps, MLOps, and cloud deployment into a single platform.

User → Career Profile → AI Resume Analysis → Skill Gap Analysis → Personalized Career Roadmap → AI Interview Preparation → Progress Dashboard



Benefits

- Integrated career-development platform
- Personalized recommendations
- AI-assisted resume and skill analysis
- Structured learning guidance
- Continuous progress tracking
- Secure and scalable architecture

TECHNICAL ASPECTS

CareerForge integrates multiple technical components to deliver an **intelligent**, **secure** and **scalable** career development platform.



FRONTEND

- React.js with TypeScript
- Tailwind CSS for styling
- Responsive UI/UX
- Components for resume upload, dashboard, and career tracking



BACKEND

- Java 21, Spring Boot 3
- RESTful APIs
- Spring Security with JWT
- Role-based access control
- Business logic and service layer implementation



DATABASE

- PostgreSQL
- Stores users, resumes, skills, recommendations, interview data, and learning progress
- Designed for scalability and data integrity



AI / ML ENGINE

- Python, Scikit-learn, Pandas
- Resume parsing and skill extraction
- Skill-gap identification
- Career recommendations
- Interview question generation



DEVOPS

- Docker containerization
- GitHub for version control
- GitHub Actions for CI/CD
- Automated build, test and deployment pipeline
- AWS for cloud infrastructure



DEVSECOPS

- SonarQube for static code analysis
- Trivy for vulnerability scanning
- OWASP security best practices
- Secure coding & dependencies management



MLOPS

- MLflow for experiment tracking
- Model versioning and registry
- Model deployment
- Lifecycle management of ML models



MONITORING

- Prometheus for metrics collection
- Grafana for visualization and dashboards
- System and application performance monitoring
- Alerts and notifications



USERS
(Students / Admins)



WEB INTERFACE
(Frontend)



APPLICATION SERVER
(Backend APIs)



DATABASE
(PostgreSQL)



AI / ML ENGINE
(Python)



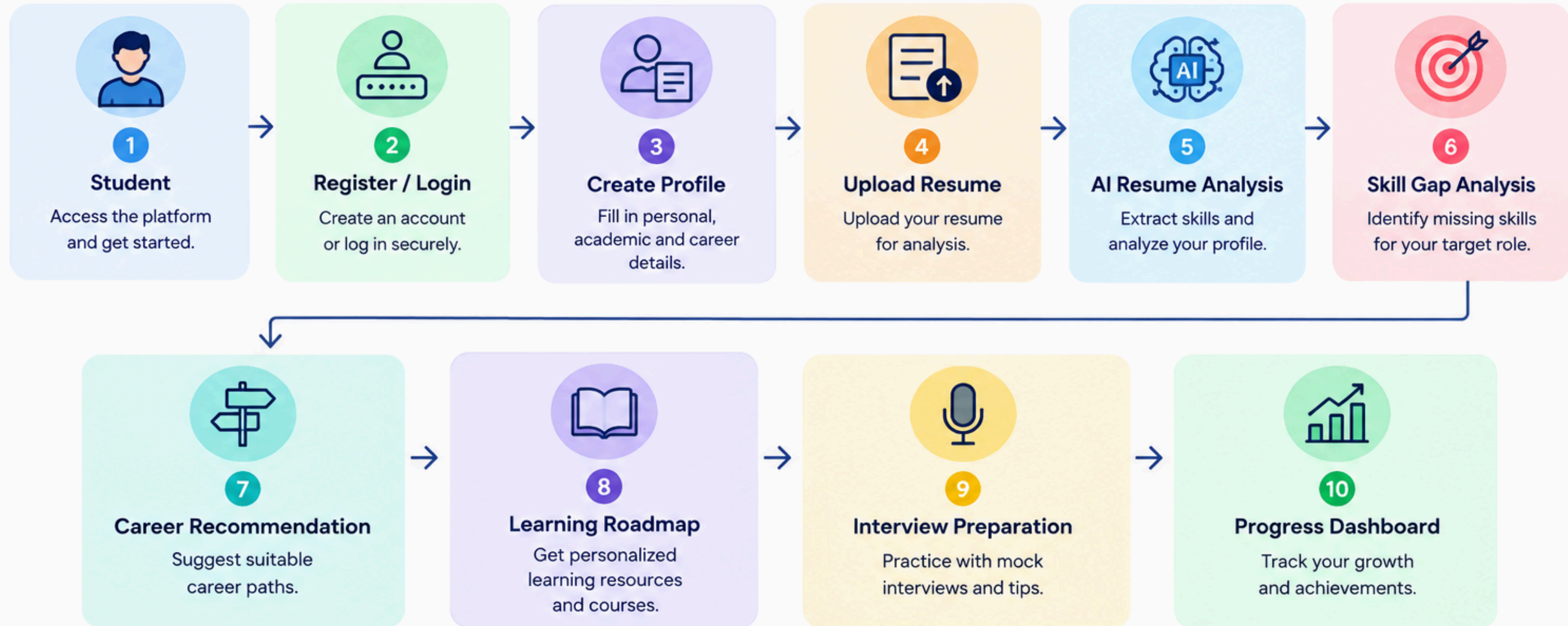
CLOUD & PIPELINE
(AWS, CI/CD, Security)



MONITORING
(Prometheus, Grafana)

CareerForge

System Workflow



Technology Stack



React.js
+ TypeScript
(Frontend)



Spring Boot
+ Java
(Backend)



PostgreSQL
(Database)



Python +
Scikit-learn + MLflow
(AI / ML)



Docker + GitHub Actions
+ AWS
(DevOps / Cloud)

Users

USERS

CareerForge has multiple user roles with distinct functionalities.



**STUDENT /
JOB SEEKER**

PRIMARY USER

-  Upload and analyze resumes
-  Identify skill gaps
-  Get personalized career recommendations
-  Practice AI-generated interview questions
-  Track career and learning progress



ADMINISTRATOR

SYSTEM MANAGER

-  Manage users and profiles
-  Manage interview questions and career resources
-  Monitor platform activity
-  View system analytics
-  Manage AI/ML-related resources



RECRUITER

FUTURE ENHANCEMENT

-  Post job/internship opportunities
-  Search candidates by skills
-  Review candidate profiles and resumes
-  Shortlist candidates



Students are the primary users of CareerForge.

Administrators manage the platform, while **recruiters** will be added as a future enhancement.



CareerForge

AI-Powered Career Development Platform

System Modules

Your Career | Our AI | A Better Tomorrow

01



Authentication & Profile

- User registration / login
- Personal, academic & career profile

02



Resume Analysis

- Resume upload
- AI-based resume and skill analysis

03



Skill Gap Analysis

- Compare current skills with target-role requirements
- Identify missing skills

04



Career Recommendation

- Suggest suitable career paths based on profile and skills

05



Personalized Learning Roadmap

- Recommend courses/resources
- Provide a structured learning path

06



Interview Preparation

- Mock interview practice
- Interview questions and preparation guidance

07



Progress Dashboard

- Track skills and learning progress
- View recommendations and achievements

Implementation

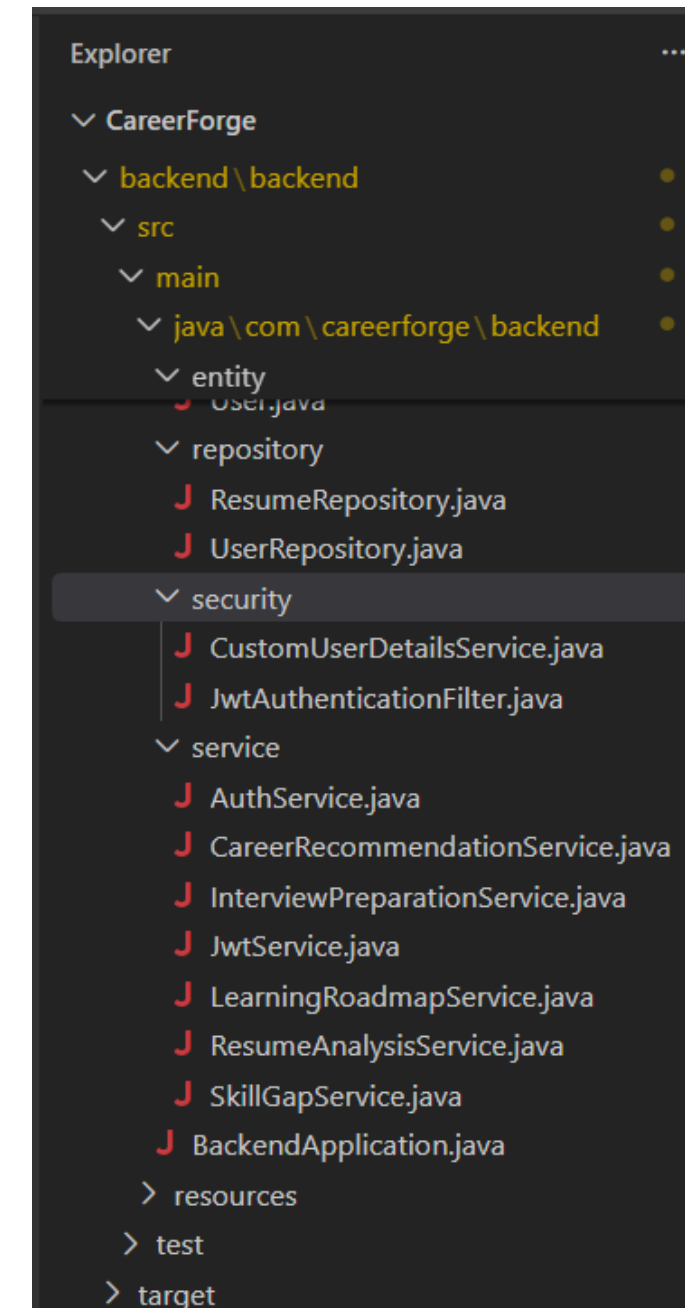
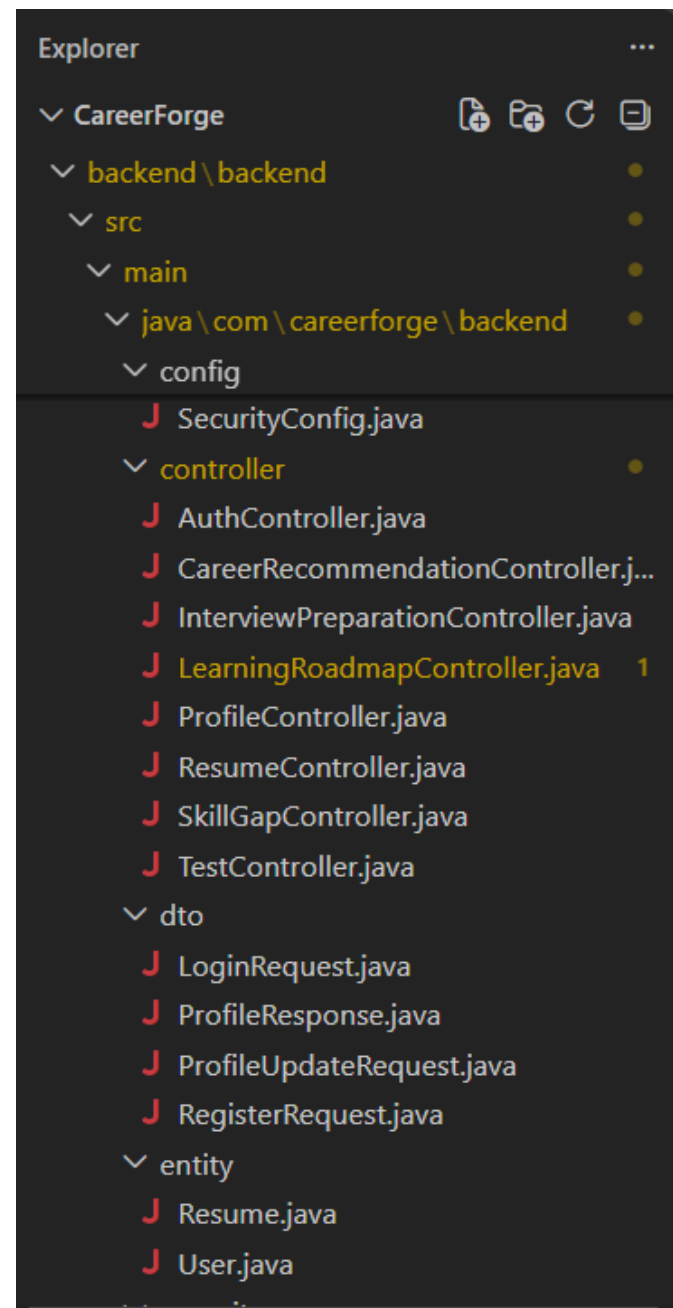
Implementation Status: 9/18 Phases Completed

Core Backend Modules Implemented

1. Authentication + JWT
2. Profile Management
3. Resume Upload & Analysis
4. Skill Gap Analysis
5. Career Recommendations
6. Learning Roadmap
7. Interview Preparation

Upcoming: React Frontend → Testing → Docker → CI/CD → DevSecOps → MLOps → Monitoring → AWS → Final Integration

Implementation



CareerForge Backend Project Structure

Implementation

```
● PS C:\Users\Vaishnavi\OneDrive\Desktop\CareerForge> curl.exe -i -X POST "http://localhost:8082/api/resume/upload" `
>> -H "Authorization: Bearer $token" `
>> -F "file=@C:\Users\Vaishnavi\Downloads\SomuVaishnavi_Resume.pdf"
HTTP/1.1 200
X-Content-Type-Options: nosniff
X-XSS-Protection: 0
Cache-Control: no-cache, no-store, max-age=0, must-revalidate
Pragma: no-cache
Expires: 0
X-Frame-Options: DENY
Content-Type: text/plain;charset=UTF-8
Content-Length: 47
Date: Tue, 22 Sep 2026 15:28:00 GMT

Resume uploaded and text extracted successfully
PS C:\Users\Vaishnavi\OneDrive\Desktop\CareerForge> Invoke-WebRequest `
```

Resume PDF uploaded and text extracted successfully

Implementation

```
PS C:\Users\Vaishnavi\OneDrive\Desktop\CareerForge> Invoke-WebRequest `
>> -Uri "http://localhost:8082/api/skill-gap" `
>> -Method GET `
>> -Headers $headers `
>> -UseBasicParsing

StatusCode      : 200
StatusDescription : 
Content         : {"missingSkills":["docker","rest api"],"targetRole":"Software Engineer","currentSkills":["java","python","javascript","react","spring boot","sql","mysql","aws","azure","git","github","machine learning...
RawContent      : HTTP/1.1 200
                  X-Content-Type-Options: nosniff
                  X-XSS-Protection: 0
                  Pragma: no-cache
                  X-Frame-Options: DENY
                  Keep-Alive: timeout=60
                  Connection: keep-alive
                  Content-Length: 493
                  Cache-Control: no-cac...

Forms           : 
Headers         : {[X-Content-Type-Options, nosniff], [X-XSS-Protection, 0], [Pragma, no-cache], [X-Frame-Options, DENY]...}
Images          : {}
InputFields     : {}
Links           : {}
ParsedHtml      : 
RawContentLength : 493
```

Skill Gap Analysis – Identifies missing skills based on the selected career role

Implementation

```
PS C:\Users\Vaishnavi\OneDrive\Desktop\CareerForge> Invoke-WebRequest `
>> -Uri "http://localhost:8082/api/learning-roadmap" `
>> -Method GET `
>> -Headers $headers `
>> -UseBasicParsing

StatusCode      : 200
StatusDescription : 
Content         : [{"duration":"1 week","topics":["Docker basics","Images and containers","Dockerfile","Docker Compose","Containerizing Spring Boot applications"],"skill":"Docker"}, {"duration":"1 week","topics":["REST ...
RawContent      : HTTP/1.1 200
                  X-Content-Type-Options: nosniff
                  X-XSS-Protection: 0
                  Pragma: no-cache
                  X-Frame-Options: DENY
                  Content-Length: 346
                  Cache-Control: no-cache, no-store, max-age=0, must-revalidate
                  Conten...

Forms           : 
Headers         : {[X-Content-Type-Options, nosniff], [X-XSS-Protection, 0], [Pragma, no-cache], [X-Frame-Options, DENY]...}
Images          : {}
InputFields     : {}
Links           : {}
ParsedHtml      : 
RawContentLength : 346
```

```
PS C:\Users\Vaishnavi\OneDrive\Desktop\CareerForge> Invoke-WebRequest `
>> -Uri "http://localhost:8082/api/career-recommendations" `
>> -Method GET `
>> -Headers $headers `
>> -UseBasicParsing

StatusCode      : 200
StatusDescription : 
Content         : [{"matchPercentage":100.0,"totalRequiredSkills":7,"career":"Software Engineer","matchedSkills":7}, {"matchPercentage":100.0,"totalRequiredSkills":4,"career":"Java Developer","matchedSkills":4}, {"matchP...
RawContent      : HTTP/1.1 200
                  X-Content-Type-Options: nosniff
                  X-XSS-Protection: 0
                  Pragma: no-cache
                  X-Frame-Options: DENY
                  Content-Length: 473
                  Cache-Control: no-cache, no-store, max-age=0, must-revalidate
                  Conten...

Forms           : 
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Images          : {}
InputFields     : {}
Links           : {}
ParsedHtml      : 
RawContentLength : 473
```

- Personalized Learning Roadmap generated from identified skill gaps
- Career recommendations based on resume skill matching

Implementation

```
PS C:\Users\Vaishnavi\OneDrive\Desktop\CareerForge\backend\backend> Invoke-WebRequest `
>> -Uri "http://localhost:8082/api/interview/questions" `
>> -Method GET `
>> -Headers $headers `
>> -UseBasicParsing

StatusCode      : 200
StatusDescription :
Content         : [{"topic":"Java","question":"What is the difference between an interface and an
                  abstract class in Java?","expectedAnswer":"Explain the main differences and
                  when each should be used."},{ "topic":"Spring...
RawContent      : HTTP/1.1 200
                  X-Content-Type-Options: nosniff
                  X-XSS-Protection: 0
                  Pragma: no-cache
                  X-Frame-Options: DENY
                  Keep-Alive: timeout=60
                  Connection: keep-alive
                  Content-Length: 982
                  Cache-Control: no-cac...
Forms           :
Headers         : {[X-Content-Type-Options, nosniff], [X-XSS-Protection, 0], [Pragma, no-cache],
                  [X-Frame-Options, DENY]...}
Images          : {}
InputFields     : {}
Links           : {}
ParsedHtml      :
RawContentLength : 982
```

Interview Preparation – Role-based interview question generation

- CareerForge provides an AI-powered platform for students to improve their career readiness.
- Integrates resume analysis, skill-gap identification, career recommendations, and interview preparation.
- Uses Agile Scrum for iterative and collaborative development.
- Implements DevOps and DevSecOps for automated, secure software delivery.
- Applies MLOps for machine-learning model management and deployment.
- Uses AWS and monitoring tools to support scalable and reliable deployment.

Conclusion

Thank You
